

AI Data Analysis Agent

End-to-End Car Sales Analysis Portfolio

Executive Summary

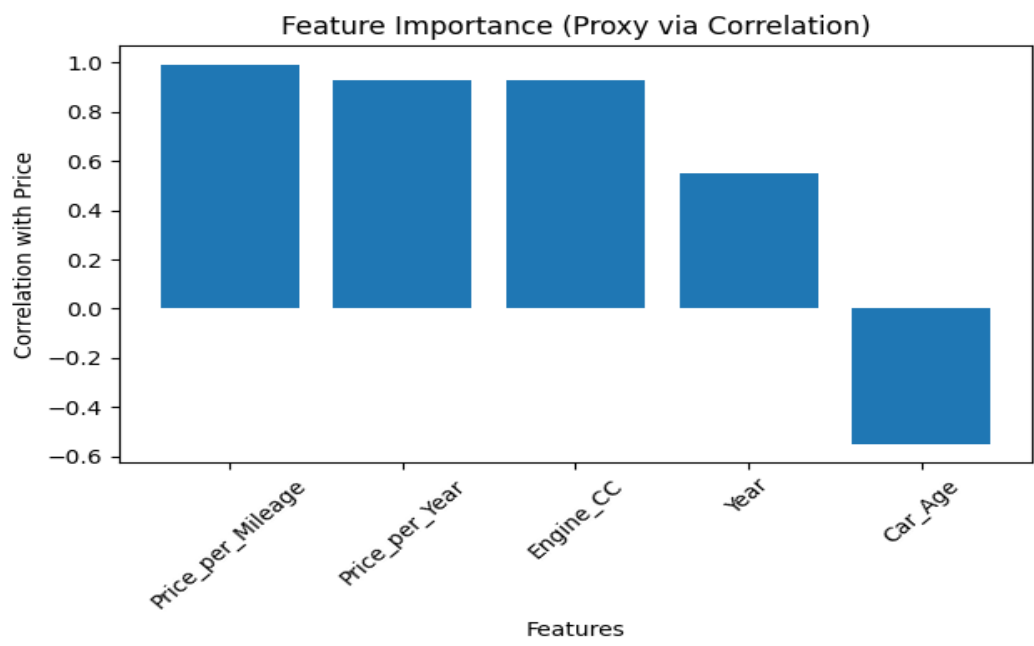
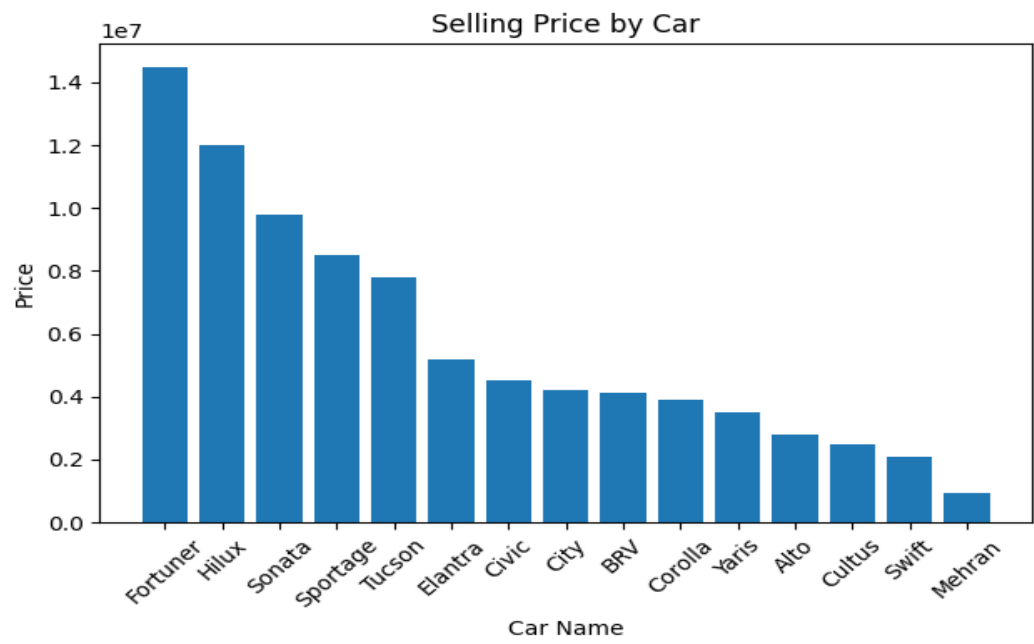
Key Insight: Vehicle pricing is primarily driven by mileage, engine size, and age, with clear segmentation between premium and budget brands.

Trends:

- Mileage negatively impacts price (depreciation curve)
- Engine size positively correlates with price
- Premium brands (Toyota/Kia) dominate higher tiers

Action: Focus pricing models on mileage and engine size for highest predictive power.

Executive Dashboard



Before vs After Cleaning

Stage	Columns
Raw Dataset	8
After Feature Engineering	11

New features added: Car_Age, Price_per_Year, Price_per_Mileage

Cleaned Dataset Sample

Company	Year	Selling_Price	Fuel_Type	Transmission	Mileage	Engine_CC	Car_Age	Price_per_Year	Price_per_Mile
Honda	2020	4500000	Petrol	Automatic	14	1800	6	750000.0	321428.57
Toyota	2019	3900000	Petrol	Manual	16	1600	7	557142.8571428572	243750.0
Honda	2021	4200000	Petrol	Automatic	15	1500	5	840000.0	280000.0
Hyundai	2022	5200000	Petrol	Automatic	13	2000	4	1300000.0	400000.0
Suzuki	2018	2100000	Petrol	Manual	18	1300	8	262500.0	116666.66
Kia	2023	8500000	Petrol	Automatic	11	2400	3	2833333.333333333	772727.27
Hyundai	2022	7800000	Petrol	Automatic	12	2000	4	1950000.0	650000.0
Toyota	2021	14500000	Diesel	Automatic	10	2800	5	2900000.0	1450000.0
Suzuki	2016	950000	Petrol	Manual	20	800	10	95000.0	47500.0
Suzuki	2023	2800000	Petrol	Manual	22	660	3	933333.3333333334	127272.72